```
[77] data = ["You are bad, you are the worst!"]
     vect = pd.DataFrame(countvect.transform(data).toarray())
     body_len = pd.DataFrame([len(data) - data.count(" ")])
     punct = pd.DataFrame([count_punct(data)])
     total_data = pd.concat([body_len,punct,vect],axis = 1)
     my_prediction = model.predict(total_data)
     print(my_prediction)

     [1]
```

```
data = ["You are amazing !!"]
vect = pd.DataFrame(countvect.transform(data).toarray())
body_len = pd.DataFrame([len(data) - data.count(" ")])
punct = pd.DataFrame([count_punct(data)])
total_data = pd.concat([body_len,punct,vect],axis = 1)
my_prediction = model.predict(total_data)
print(my_prediction)

[0]
```

Figure 6: Sample predictions made

```
model = model.fit(X_train,y_train)
model.score(X_test,y_test)

0.6190476190476191
```

The following piece of code uses the stochastic gradient descent algorithm. The output shows that we got an accuracy of around 49 per cent.

```
from sklearn.linear_model import
SGDClassifier
model = SGDClassifier()
```

```
model = model.fit(X_train,y_train)
model.score(X_test,y_test)

0.49206349206349204
```

The following piece of code uses Naive Bayes. We get an accuracy of around 60 per cent.

```
from sklearn.naive_bayes import
GaussianNB
model = GaussianNB()
model.fit(X_train, y_train)
model.score(X_test,y_test)

0.6009070294784581
```

Now that we have checked out all the algorithms, let us graph their accuracy performance. The graph is shown in Figure 5.

As you can see, the random forest algorithm gave the best accuracy for this problem and we can conclude that it is the best fit for sentiment analysis amongst ML algorithms. We can improve the accuracy much more by getting better features, trying out other vectorising techniques, and using a better data set or newer classification algorithms.

Now that random forest is seen as the best algorithm for this problem, I am going to show you a sample prediction. In Figure 6, you can see that the right predictions are being made! Do try this out to improve upon this project! END

By: Jishnu Saurav Mittapalli

The author is a full stack Web developer, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.